Q.16 How do you verify connectivity and monitor a network after configuration? (CO3)

Q.17 Briefly describe the address Resolution Protocol (ARP) process. (CO2)

Q.18 Define the role of Device Interconnections in network planning. (CO2)

SECTION-D

Note: Long answer questions. Attempt any one question out of two questions. (1x10=10)

Q.19 Discuss the full cycle of planning a network, from physical connections and device interconnections to developing an addressing scheme and calculating subnets. (CO4)

Q.20 Discuss Ethernet in detail, including the Physical Layer, MAC, frames, and the steps required to verify and test connectivity in a new network. (CO4)

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Roll No. ....

2nd Sem. / DVOC (Level-4) / SD

Subject : Maintenance of Computer Systems

Time : 2Hrs. M.M. : 50

SECTION-A

Note: Very short Answer Questions. Attempt all questions. (5x1=5)

Q.1 Which layer of the OSI model is responsible for representing bits as signals? (CO1)

a) Data Link Layer b) Network Layer

c) Physical Layer d) Transport Layer

Q.2 In an Ethernet network, which sub-layer is responsible for hardware addressing? (CO1)

a) LLC b) MAC

c) IP d) ARP

Q.3 Which protocol maps a known IP address to a MAC address on a local network? (CO1)

a) DHCP b) ARP

c) DNS d) HTTP

Q.4 A device that forwards data only to a specific destination port is a: (CO1)

a) Hub b) Switch

c) Repeater d) Gateway

Q.5 What is a primary task when "Planning and Cabling Networks"? (CO1)

a) Developing an addressing Scheme

b) Writing software drivers

c) Designing a logo

d) Installing an OS

SECTION-B

Note: Objective/Completion type Questions. All questions are compulsory. (5x1=5)

Q.6 The process of converting bits into signals is called _________. (CO1)

Q.7 Determining how many networks are needed involves _______ the subnets. (CO1)

Q.8 Twisted pair and fiber optics are examples of _______ used to connect devices. (CO1)

Q.9 To verify connectivity between two devices, the _______ command is most commonly used. (CO1)

Q.10 Ethernet operates at both the Data Link layer and the _______ layer. (CO1)

SECTION-C

Note: Short Answer Questions. Attempt six questions out of eight questions. (6x5=30)

Q.11 Describe the role of the Physical Layer in signal transmission. (CO2)

Q.12 Differentiate between Hub and Switch. (CO2)

Q.13 Outline the basic structure and components of an Ethernet Frame. (CO2)

Q.14 What factors are essential when planning LAN physical connections? (CO2)

Q.15 Explain the importance of an Addressing Scheme in network configuration? (CO3)